

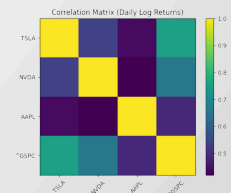


The Tech Domino Effect: Volatility Transmission from Mega-Caps to the S&P 500

Vo Thi Quynh Tram, Lai Chiu Lok, Huei-Wen Teng

Motivation

- Rising dominance of mega-cap tech firms in the market.
- Potential volatility spillovers affecting overall market movements.
- Need to assess systemic risk from concentrated market influence.



Outline

1. Motivation

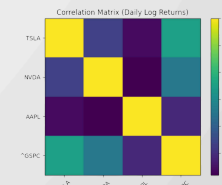


2. Data

3. Multiple Linear Regression

4. Results

5. Conclusion



Data Preprocess_Data Overview

Period: 2015–2024 (Daily)

Source: Yahoo Finance (Adjusted Close)

Assets analyzed:

Apple (AAPL)

Nvidia (NVDA)

Tesla (TSLA)

S&P 500 (Benchmark)

Transformation: Log returns $\rightarrow$ Rolling Volatility (annualized)



Data Preprocess_Variables

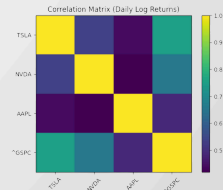
Variable	Definition	Role
RV_SPX	Rolling Volatility of S&P 500	Dependent
RET_AAPL, RET_NVDA, RET_TSLA	Log Returns of tech firms	Independent
RV_AAPL, RV_NVDA, RV_TSLA	Rolling Vol of tech firms	Spillover Drivers



Data Preprocess_Descriptive Statistics

Ticker	count	mean	std	min	25%	50%	75%	max	Skew	Kurtosis	JB_pvalue
AAPL	2514	0.00093	0.017934	-0.137708	-0.007355	0.001	0.010093	0.113158	-0.201228	5.457399	0
NVDA	2514	0.002248	0.030378	-0.207712	-0.012514	0.002648	0.017472	0.260876	0.208734	6.829886	0
TSLA	2514	0.001333	0.035911	-0.236518	-0.016292	0.001264	0.019073	0.198187	-0.066409	4.410539	0
SPX	2514	0.000419	0.011272	-0.127652	-0.00378	0.000639	0.00574	0.089683	-0.81083	15.750819	0

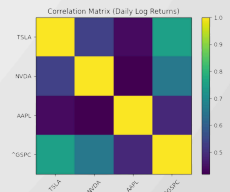
- All assets: Mean ≈ 0 , high variance (Std highest for TSLA)
- Evidence of non-normality:
 - Positive/negative skew $\rightarrow$ asymmetric risk
 - High kurtosis $\rightarrow$ heavy tails
 - JB-test $\rightarrow$ rejects normality ($p = 0$)



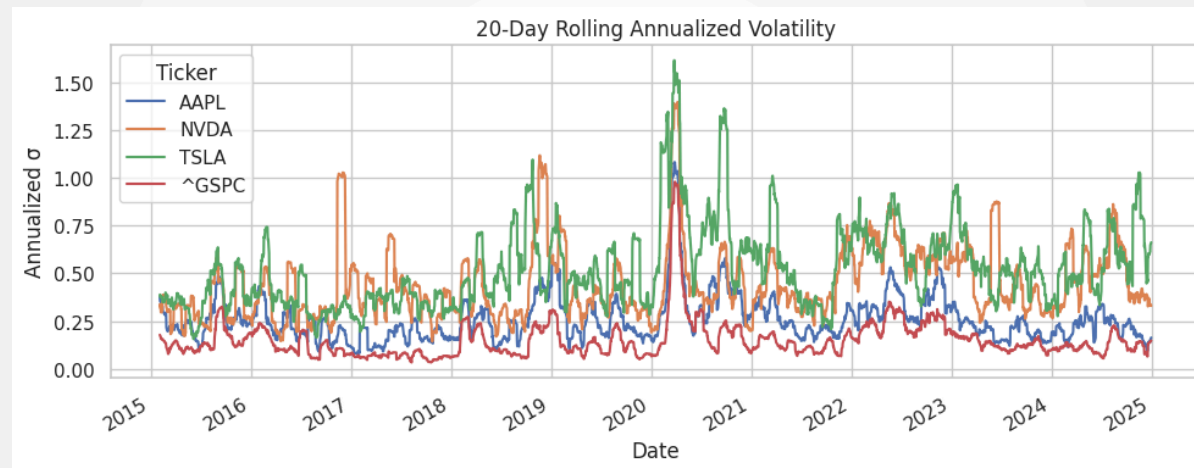
Data Preprocess_Extreme Events ($|z| > 3$)

Asset	Count
TSLA	40
AAPL	40
NVDA	29
S&P500	36

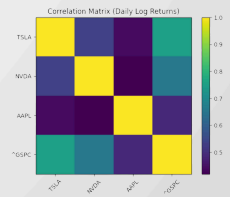
TSLA & AAPL most prone to tail shocks → candidates for volatility domino effects.



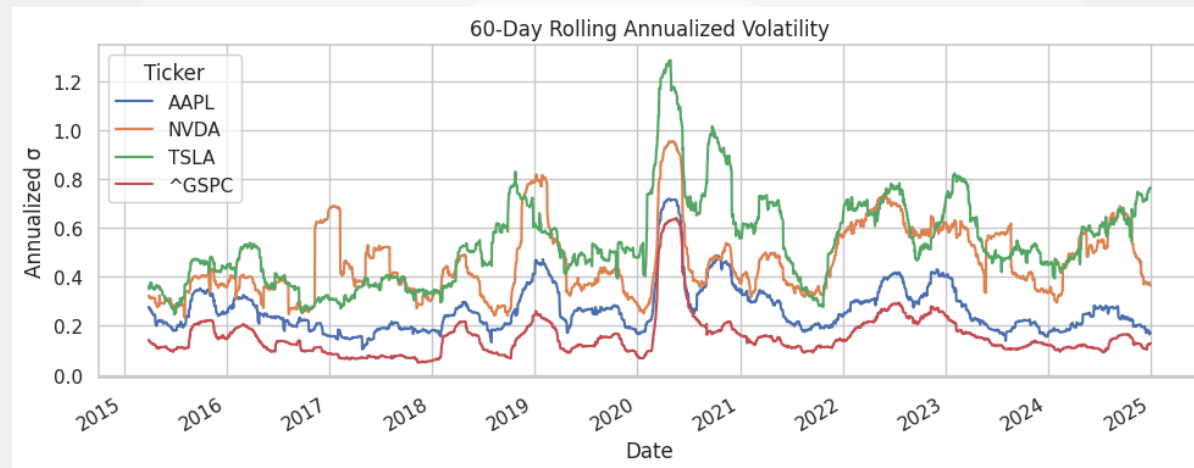
Data Preprocess_Rolling Volatility



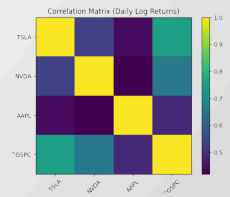
- TSLA & NVDA show sharp volatility spikes (2020–2021)
- SPX volatility rises after mega-cap spikes → Possible contagion
- AAPL = moderate, smoothed volatility (market stabilizer)



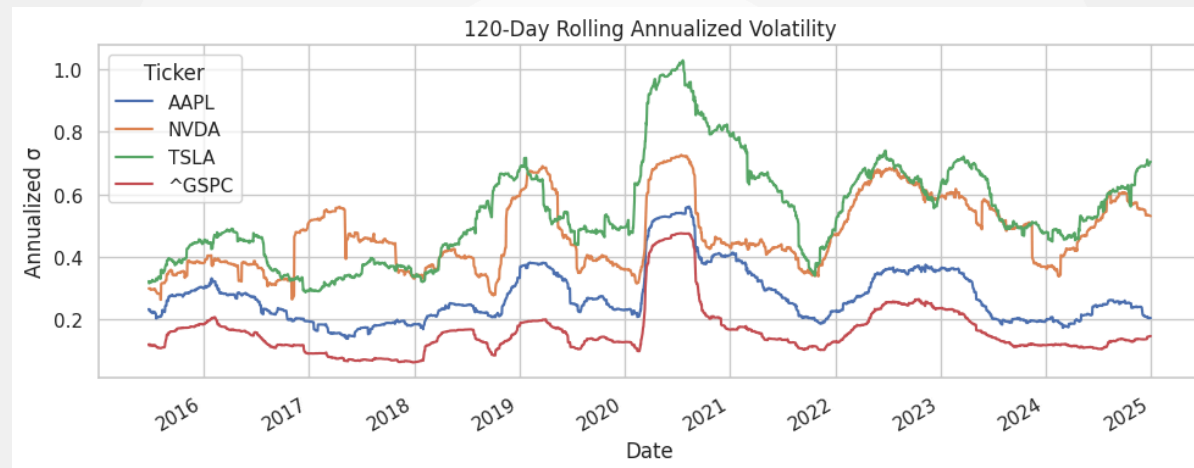
Data Preprocess_Rolling Volatility



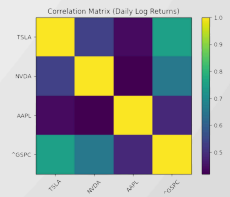
- TSLA & NVDA show sharp volatility spikes (2020–2021)
- SPX volatility rises after mega-cap spikes → Possible contagion
- AAPL = moderate, smoothed volatility (market stabilizer)



Data Preprocess_Rolling Volatility



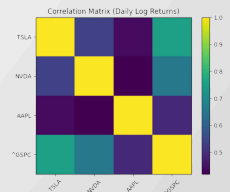
- TSLA & NVDA show sharp volatility spikes (2020–2021)
- SPX volatility rises after mega-cap spikes → Possible contagion
- AAPL = moderate, smoothed volatility (market stabilizer)



Data Preprocess_Market Sensitivity

Asset	Beta vs S&P500
NVDA	1.73
TSLA	1.5
AAPL	1.19

- NVDA & TSLA amplify market risk (“leveraged exposure”)
- High Beta supports volatility transmission hypothesis



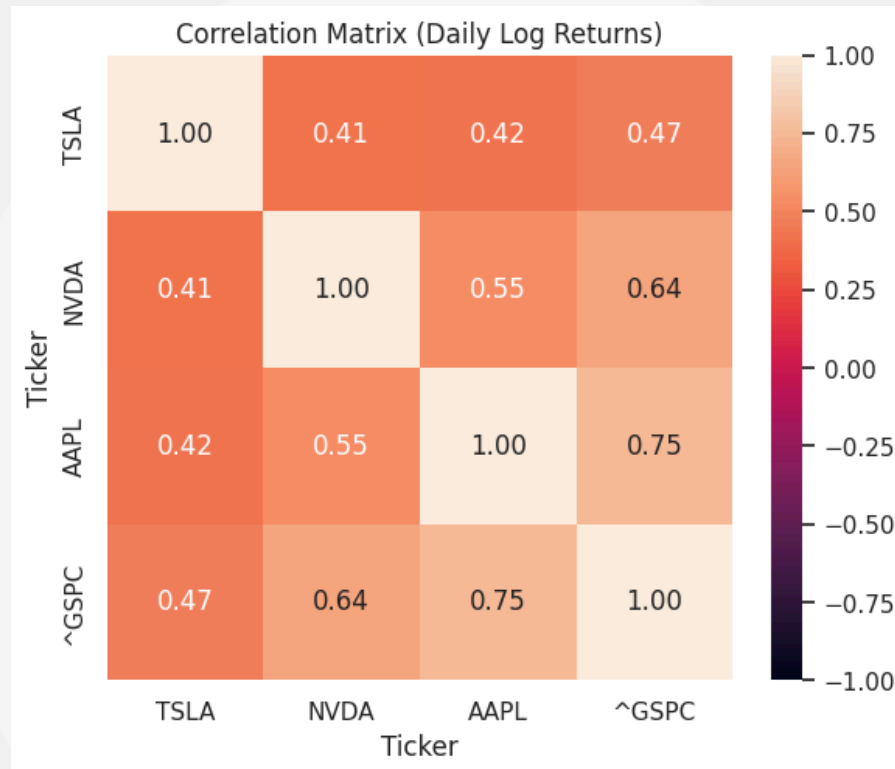
Data Preprocess_Risk Metrics (MaxDD,CVaR)

Asset	MaxDD	CVaR(95%)
TSLA	-73.60%	-0.083
NVDA	-66.30%	-0.067
AAPL	-38.50%	-0.042
S&P500	-33.90%	-0.028

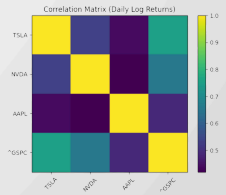
- Mega-caps have deeper crash risk → Potential systemic propagation.



Data Preprocess_Correlation & Co-Movement



- AAPL highly correlated with SPX (~0.75)
- TSLA shows episodic decoupling (idiosyncratic risk)
- NVDA moderately correlated, rising in AI boom
- Conclusion: Strong integration but not perfect coupling → transmission, not redundancy.



Data Preprocess_Key Findings (Volatility Transmission)

- Evidence of fat tails, extreme risk, and high Beta in mega-caps
- S&P500 volatility spikes lag mega-cap volatility surges
- Tech sector acts as a volatility leader, not follower

